

LARW: Robust Watermarking Framework With Linear Additive Residuals

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Abstract—The deep learning-based watermarking approach has emerged as a leading technique within image digital watermarking, demonstrating strong capabilities in generation quality, robustness, and generalization. However, most existing watermarking methods rely on the encoder–noise layer–decoder (E–N–D) structure, which presents certain limitations; specifically, it is challenging to simultaneously enhance generation quality, robustness, and generalization performance. To address these limitations, this paper introduces a novel robust watermarking framework with linear additive residuals (LARW), which eliminates the traditional intermediate noise layer. Instead, it directly produces a robust residual through a redesigned encoder with robust residuals (RRE). Specifically, we introduce an innovative encoder structure RRE, which allows each noise type to be processed in parallel, generating its respective robust residual. Through linear additivity, these residuals are combined to produce a strong, robust residual capable of resisting multiple noise types. By adding residual to the original image, we obtain the watermarked image, which can then be input directly into the decoder for watermark extraction, bypassing the need for image distortion simulation. Additionally, both encoder and decoder employ convolutional and deconvolutional architectures, respectively, ensuring that the decoder closely approximates the encoding space during watermarked image decoding. This alignment facilitates rapid watermark synchronization and decoupling of watermark features. Experimental results demonstrate that the proposed method achieves an improved balance between imperceptibility and robustness with an average PSNR exceeding 40.70 dB, SSIM exceeding 0.99 and an average accuracy exceeding 95% under an embedding strength of 1.0. The results show that the proposed method outperforms existing schemes in terms of

robustness under JPEG compression, Cropout, Dropout, Gaussian noise, Gaussian filter, and Scaling.

Index Terms—Robust watermarking, robust residual, linear additivity.

I. INTRODUCTION

WITH the development of artificial intelligence, the need for copyright protection and traceability of multimedia content is becoming increasingly critical [1], [2], [3], [4], [5], [6]. For example, copyright protection of images, text, audio, video and other media can effectively prevent data misuse and mitigate risks related to public opinion. Particularly in the era of large-scale content generation by AI models, various countries have established standards to mandate the identification of AI-generated content, enabling authentication or traceability of such material [7], [8], [9], [10]. Copyright protection and traceability remain paramount security concerns, as determining data ownership is essential in managing copyright disputes, whether concerning original or AI-generated content. Digital watermarking [11], [12], [13] has proven to be a highly effective solution to these issues. By embedding specific identifiers or watermarks within carrier data, digital watermarking ensures that the content remains imperceptible to human observers, meaning the watermarked data retains invisibility. Additionally, digital watermarking is designed to resist certain types of attacks, allowing successful extraction of watermark even when the watermarked data has been compromised. Current digital watermarking techniques can withstand standard attack types, including JPEG compression, Gaussian noise, and average filtering. However, when faced with complex real-world application scenarios or more malicious attacks, digital watermarking still encounters limitations in terms of visual quality of the watermarked image, robustness, and embedding capacity. Therefore, developing digital watermarking methods that achieve both high robustness and superior visual quality remains a significant scientific challenge.

Digital watermarking technology has seen substantial advancements from traditional methods to deep learning-based approaches, with improvements across generation quality, robustness, embedding capacity, and embedding efficiency. Traditional watermarking methods are primarily grounded in cryptography, statistics, and other theoretical principles, using manually designed algorithms to embed and extract watermarks [14], [15], [16]. These methods are theoretically robust, straightforward to implement, easy to understand, and have relatively low computational requirements. Common approaches can be categorized

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into spatial domain and transform domain methods [17], [18]. Spatial domain methods directly modify pixel values within the spatial domain to embed watermarks, with the least significant bit (LSB) technique being a typical example [19]. In contrast, transform domain methods embed watermarks by altering coefficient values in the frequency domain, utilizing techniques such as Discrete Cosine Transform (DCT) [20], [21], Discrete Wavelet Transform (DWT) [22], [23], [24], Discrete Fourier Transform (DFT) [25], [26] and so on. Research indicates that embedding watermarks in the transform domain generally enhances robustness and improves visual quality. Different transform domains exhibit varying levels of resistance to specific attacks; for instance, DCT-based methods are more resilient to JPEG compression, as JPEG compression operates on DCT principles, while DFT-based methods demonstrate higher resistance to rotation due to the rotational invariance of DFT transformations. The mature theoretical knowledge of traditional watermarking methods provides an important theoretical basis for in-depth research on deep learning-based watermarking techniques.

Currently, numerous deep learning watermarking methods have been developed, with several prominent examples [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40]. Zhu et al. [28] first proposed a comprehensive end-to-end watermarking framework, HiDDeN, which consists of an encoder, noise layer, decoder, and discriminator. The encoder embeds the watermark into the carrier image, the noise layer simulates image distortion, the decoder extracts watermark information from the distorted image, and the discriminator aims to distinguish between the carrier and watermarked images, thereby enhancing network performance. Tancik et al. [29] introduced Stegastamp, a U-Net-based [41] watermarking framework that improves robustness by optimizing the embedding position within the encoder and noise layer simulation. Huang et al. [32] proposed an attention-based robust image watermarking model, ARWGAN, leveraging a generative adversarial network (GAN) [42]. Through a feature fusion and attention module, ARWGAN achieves multi-layer feature extraction and adaptive watermark embedding, effectively increasing robustness. To address issues in gradient backpropagation caused by noise layers, Jia et al. [30] introduced a mini-batch strategy, which randomly selects from Identity, real JPEG, and analog JPEG noise layers in each batch. This approach leverages optimizer characteristics to maintain consistent gradient backpropagation, thereby enhancing robustness. Ge et al. [33] improved robustness by combining multiple noise types for image distortion simulation via serial connection, although the resulting watermarked images still show limitations in visual quality. To further enhance both robustness and visual quality, Fang et al. [34] developed a decoder-driven watermark framework, DeEND, which improves feature coupling between the encoder and decoder. Despite its effectiveness, DeEND's reliance on training with a single noise type limits its adaptability. In efforts to advance deep learning watermarking, researchers have focused on optimizing network structure, noise layers, embedding strength, attention modules, and related components. For instance, Ma et al. [31] proposed

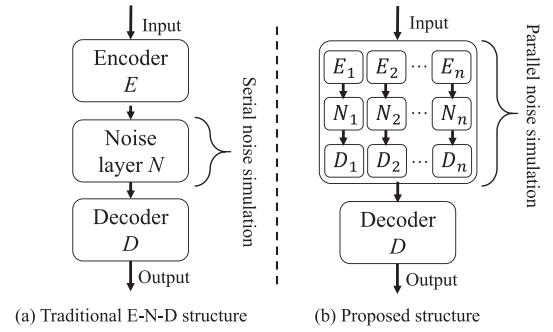


Fig. 1. The difference between the traditional structure and the proposed structure.

CIN, a framework combining reversible and irreversible mechanisms. CIN's performance is enhanced through the inclusion of diffusion and extraction modules, reversible modules, fusion and segmentation modules, an irreversible attention module, and tailored noise selection modules. Fang et al. [43] introduced DeNoL, a cross-channel robust network featuring a decoupling noise layer for cross-channel simulation, which requires only a few samples for training. Qin et al. [44] proposed NSN, a network that simulates complex noise fusion using a multi-task loss function based on the just-noticeable-difference (JND), significantly reducing watermarked image distortion. Wang et al. [45] proposed a separation-trained adaptive watermarking method, where embedding strength is controlled through adaptors, enabling an automatic intensity factor that further enhances visual quality and robustness. Ma et al. [35] advanced a new model architecture utilizing Swin-Transformer [46] and deformable Convolutional Network (DCN) [47], significantly boosting robustness against geometric distortions by initially training an image distortion style conversion network, followed by joint encoder-decoder training. From existing deep learning-based watermarking methods, it is not difficult to observe that their network architecture essentially adopts an encoder-noise layer-decoder structure (E-N-D), as illustrated in Fig. 1(a). The differences lie in the fact that some methods may employ two-stage separate training or simulate distortions in the noise layer in advance. In addition, the designs of the encoder and decoder architectures also differ. Beyond using multiple convolutional layers for feature extraction, extra modules are often incorporated, such as wavelet decomposition to replace the downsampling process, watermark coupling schemes, self-attention modules, and so on. The design of these modules plays a certain role in improving visual quality and robustness.

Although deep learning watermarking methods achieve enhanced performance, they also present challenges, such as fixed-resolution image input requirements, limited theoretical interpretability in model structure design, lower visual quality of generated images, and insufficient generalization and robustness. Additionally, many existing methods [28], [29], [30], [31], [32], [33], [34], [35] employ the E-N-D structure, wherein robustness is primarily derived from the noise layer. The noise layer typically enhances robustness by simulating either a single noise type or a sequence of multiple noise types. Although

this method improves robustness to a certain extent, it employs serial noise simulation by sequentially combining multiple noise types, which essentially mimics a strong combined attack. However, this strategy makes it difficult for the network to optimize robustness against each individual noise type simultaneously. Therefore, designing a novel network architecture that can effectively optimize robustness across various noise types while ensuring high visual quality of the watermarked images remains a valuable and meaningful research challenge.

To address these challenges, we propose a robust watermarking framework with linear additive residuals (LARW). By analyzing the linear additivity of noise in complex scenarios, the proposed framework effectively separates different types of noise and optimizes robustness against each noise type in a parallel manner. This framework removes the intermediate noise layer, instead directly generating robust residuals through a redesigned encoder with robust residuals (RRE). These residuals are then added to the original image, producing the watermarked image. Furthermore, encoder and decoder adopt convolutional and transposed convolutional structures, ensuring that the decoder can approximately reconstruct the encoding space of the encoder, thereby achieving synchronization of watermark features and effectively decoupling them. Comprehensive experimental evaluations based on multiple performance metrics demonstrate that the proposed method achieves significant advantages over existing methods. The main contributions of the paper are as follows:

- We propose a novel robust watermarking framework, LARW, which eliminates the intermediate noise layer and directly generates robust residuals through a redesigned encoder RRE.
- We provide a theoretical analysis of various noise types using the principle of linear additivity and incorporate them in parallel within the encoder RRE to enhance robustness.
- Experimental results demonstrate that the proposed method achieves superior performance in terms of robustness and generalization, as well as visual quality metrics.

The rest of this paper is organized as follows. In Section II, we review deep learning watermarking methods and analyze the theory accordingly. In Section III, we introduce the LARW watermarking framework proposed, which mainly includes encoder, decoder and loss function. Section IV is the experimental results and analysis. The last part is the summary and prospect.

II. REVIEW AND ANALYSIS

A. Brief Review

In considering deep learning watermarking method, we conduct further analysis of the E-N-D structure. It is easy to know that for a trained watermarking network, let the encoder, noise layer and decoder be denoted by E , N and D , respectively. Given any original image $X \in \mathbb{R}^{m \times n \times c}$ and watermark $W \in \{0, 1\}^L$, its watermarked image can be represented as $X_{Emb} = E(X, W)$, the noised image as $X_{Noise} = N(X_{Emb})$, and the extracted watermark as $W_{Ext} = D(X_{Noise})$. In addition, the performance of encoder and decoder is mainly measured by the visual quality of the generated watermarked image and the

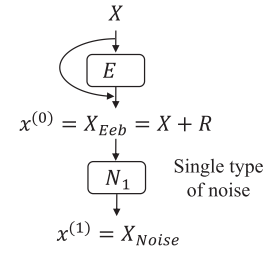


Fig. 2. Encoder E and noise layer N with a single type of noise. The encoder E generates watermarked image X_{Emb} , which is then fed into noise layer N_1 with a single type of noise for image distortion simulation to improve the robustness of X_{Emb} .

accuracy of watermark extraction. In the encoder, the generated watermarked image X_{Emb} should remain visually imperceptible from the original image X . In the decoder, higher extraction accuracy for W_{Ext} indicates better performance. Therefore, the optimization process of encoder and decoder is a contradictory and antagonistic process. Optimizing the performance of codecs simultaneously is a difficult and challenging task.

In this process, we define the residual between the watermarked image X_{Emb} and the original image X as $R = X_{Emb} - X$. Then, for the encoder E , the visual quality of the generated watermarked image X_{Emb} depends primarily on the location and intensity of pixel modifications, which are, in turn, determined by the residual R . Consequently, for any structure of the encoder E , the outcome of their learning is the residual R . Meanwhile, the extraction accuracy of the watermark W_{Ext} (i.e., robustness) depends significantly on the noise layer N . Since the encoder E , the noise layer N , and the decoder D are trained jointly, the main response for robustness is on the residual R . Through simulating noise in layer N , the network optimizes R to achieve resilience against distortions. Therefore, developing a robust residual R that balances visual quality and robustness is the central challenge in improving overall watermarking performance.

B. Separability Analysis

As robustness in watermarking networks primarily depends on the noise layer N , adding specific noise types to N can target robustness against those particular distortions. Existing approaches often train networks by either introducing single noise types or serially connecting multiple noise types, as illustrated in Figs. 2 and 3. While this approach offers resistance against various noise attacks, we posit that the residual R produced through this design may not achieve optimal robustness. Considering the linear additivity of noise, we assume that the residual R is separable. Let $N_i, i = 1, 2, \dots, n$ represent distinct noise types and its corresponding residual as $R_i, i = 1, 2, \dots, n$. Then the relationship between the watermarked image X_{Emb} and the original image X can be expressed as follows:

$$X_{Emb} = X + R = X + \sum_{i=1}^n \omega_i R_i \quad (1)$$

where ω_i is a weight, constrained by $0 \leq \omega_i \leq 1$ and $\sum_{i=1}^n \omega_i = 1$, whose value will be analyzed further. Eq (1) shows that the

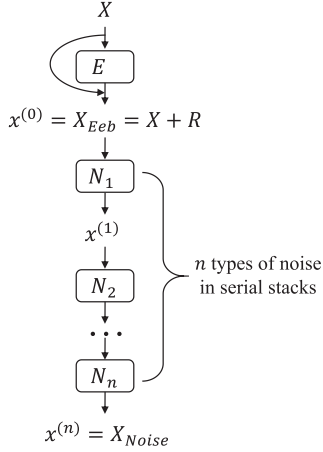


Fig. 3. Encoder E and noise layer N with multiple types of noise stacked serially. The encoder E generates watermarked image X_{Emb} , which is then fed into noise layer N with multiple types of noise series for image distortion simulation to improve the robustness of X_{Emb} .

residual R obtained by network learning is learned by the network is a linear combination of multiple noise types, providing a theoretical foundation for designing the network.

In addition, according to $X_{Emb} = E(X, W)$, we can rewrite this as:

$$X_{Emb} = E(X, W) = E(X) + E(R, W) = X + E(R) \quad (2)$$

Equation (2) indicates that encoder E needs only to learn the residual R . According to (1) and the linear additivity of noise, we further transform (2) as:

$$X_{Emb} = X + E(R) = X + \sum_{i=1}^n \omega_i E(R_i) \quad (3)$$

Equation (3) reveals the fact that the residual R can be learned by the encoder E independently, and then obtained by linear sum of all noise types. Therefore, when designing the encoder E , the residual R is directly related to the design of various types of noise. How to design various types of noise directly affects the robustness of residual R , and determines the upper limit of network robustness. Based on this theoretical analysis, we propose an innovative encoder structure, which called encoder with robust residuals (RRE). The structures of RRE are shown in Fig. 4. And the RRE diverges from prior watermarking networks in two key aspects:

- **Encoder Design:** The RRE includes simulations of multiple noise types directly within the encoder to derive robust residuals for each type of noise, unlike prior methods where the encoder outputs residuals that then input into a noise layer.
- **Residual Optimization:** The residual R in RRE is theoretically optimized for robustness, as opposed to traditional encoders where robustness derives solely from the noise layer N , lacking a theoretical foundation.

According to the theoretical analysis, the proposed framework removes the need for the noise layer N by generating a robust residual R directly. The residual R is added to the original image X to generate watermarked image X_{Emb} , which is then fed to

the decoder D for watermark extraction. In addition, we need to discuss the weight ω_i , because the weight ω_i is related to the robustness of the residual R . According to formula (3), the robustness of residual R is a composite of multiple noise types, directly tied to watermark extraction accuracy. Therefore, the value of weight ω_i can be defined as:

$$\omega_i = [\text{softmax}(\omega)]_i = \frac{\exp(\omega_i)}{\sum_j \exp(\omega_j)} = \frac{\exp(1 - \text{Acc}_i)}{\sum_j \exp(1 - \text{Acc}_j)} \quad (4)$$

where Acc_i is the watermark extraction accuracy of decoder D under noise attack type N_i . Eq (4) reflects the proportion of the robust residuals R_i of the current noise attack type N_i . In the experiment, watermark cross-entropy could substitute for Acc_i , as it also correlates with robustness, and can serve as an optimizable loss function, further enhancing noise robustness.

III. METHODS

A. Overall Architecture

Based on the theoretical analysis in Section II, we propose a robust watermarking framework with linear additive residuals (LARW), depicted in Fig. 5. This framework primarily consists of two components: an encoder RRE and a decoder D . In the encoder RRE, a robust residual R that resists multiple noise types is learned and summed with the original image X to generate the watermarked image X_{Emb} . This watermarked image X_{Emb} is subsequently passed to the decoder D , which extracts the watermark W_{Ext} . Where the robust residuals R is linearly additive, which is optimized by multiple noises in parallel, and weighted by calculating the accuracy of watermark extraction. Our framework removes the noise layer N in E-N-D structure and generates the robust residual R directly, thus obtaining a highly resilient watermarked image X_{Emb} . Below, we provide detailed descriptions of the encoder, decoder, and loss function design within the framework LARW.

B. Encoder RRE

From the theoretical analysis in Section II, we know that the result of the encoder RRE learning is essentially residual R , which directly influences both visual quality and robustness. From the perspective of visual quality, we hope that the smaller the residual R , the better, because the obtained watermarked image X_{Emb} has a smaller pixel amplitude modification and is closer to the original image X . From the perspective of robustness, we hope that the residual R can be robust. Simply put, the value of the residual R is large, and even if there is a noise attack, it does not affect the watermark value at each position represented by the residual R (if the watermark is 0 or 1 b, it is equivalent to the watermark has not flipped).

In order to learn residuals R with robustness, the encoder RRE learns residuals R for each noise type N_i in parallel. Specifically, for noise type N_i , we first use encoder E_i learning to get residual R_i :

$$R_i = E_i(X, W) \quad (5)$$

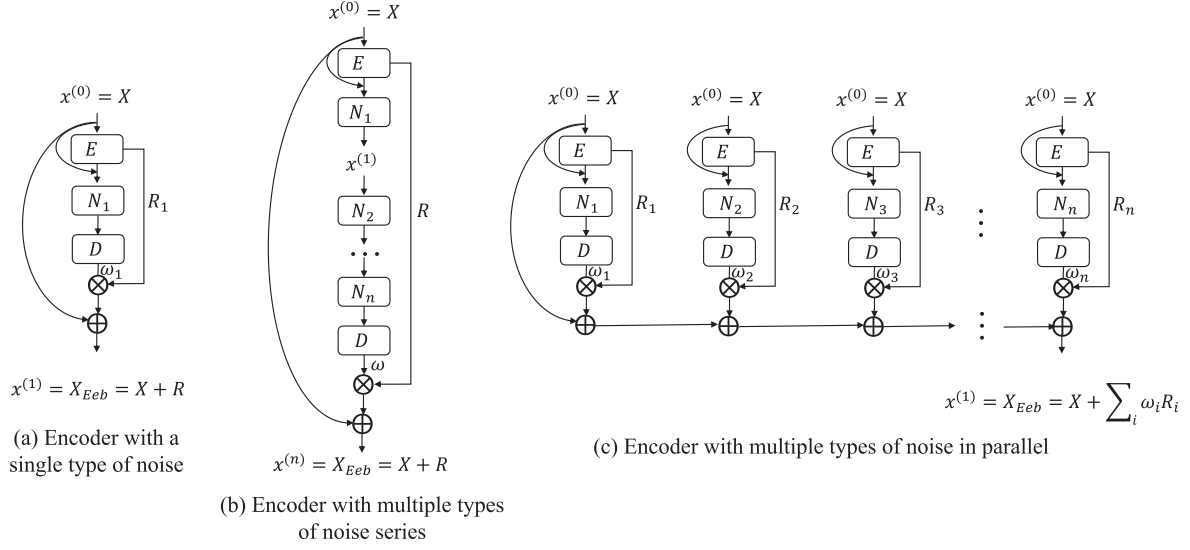


Fig. 4. The encoder RRE structures. (a) illustrates an encoder designed to handle a single type of noise, generating a robust residual R_1 corresponding to noise type N_1 . In (b), the encoder incorporates multiple types of noise in a serial arrangement, enabling the generation of a robust residual R that accounts for multiple noise combinations. (c) presents an encoder with multiple noise types arranged in parallel, producing a separable robust residual R obtained as a linear combination of individual robust residuals R_i ($i = 1, 2, \dots, n$) corresponding to each type of noise.

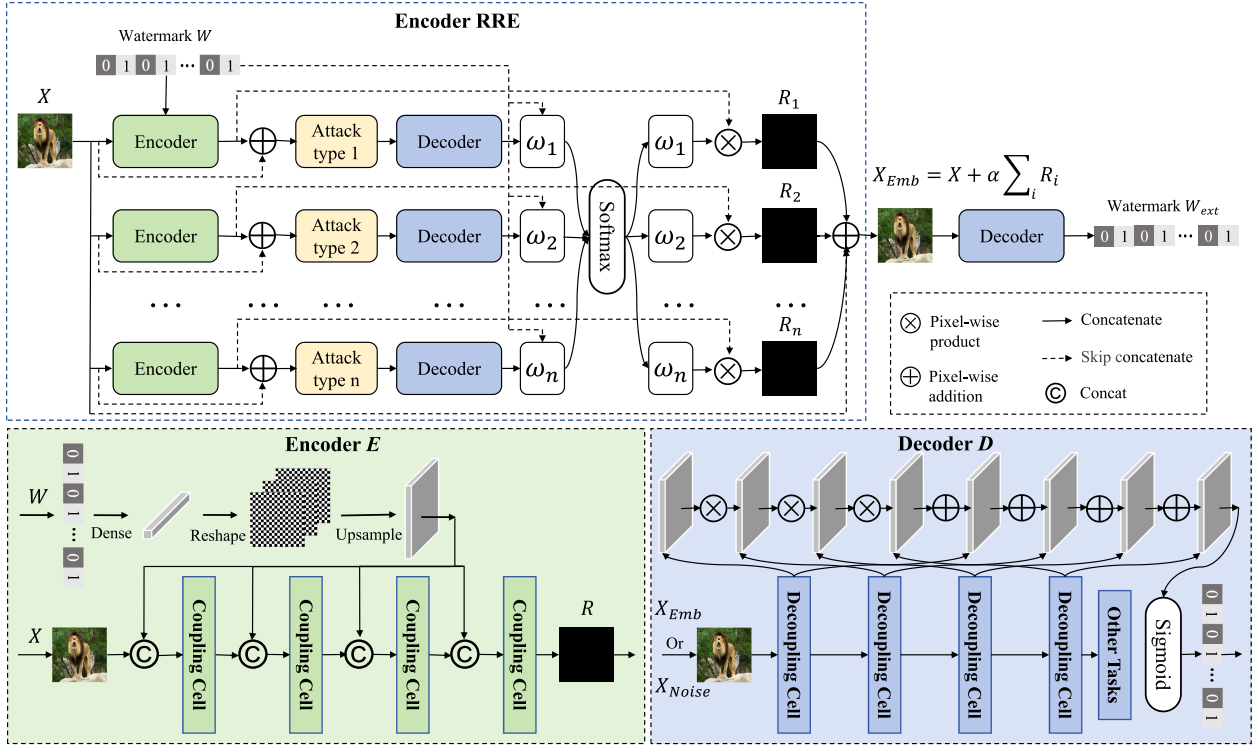


Fig. 5. The architecture of the proposed watermarking framework (LARW). The LARW framework generates a robust residual R directly from the encoder RRE and adds it to the original image X to produce the watermarked image X_{Emb} . The framework removes the noise layer N from the E-N-D structure, and feeds the watermarked image X_{Emb} directly into the decoder D for watermark extraction, without the need to simulate image distortion.

and then improve its robustness by attacking noise type N_i . To make the overall residual R robust for all noise types, we need to add a decoder D_i to learn the robustness weights ω_i for noise type N_i :

$$\omega_i = \mathcal{L}(W, D_i(R_i + X)) \quad (6)$$

where $\mathcal{L}$ is the cross-entropy function. Then, for noise type N_i , its ω_i is represented by watermark cross-entropy and added to the overall loss function for optimization. At the same time, in order to optimize the visual quality, we need to conduct overall constraints on residual R , so that it has strong robustness and small pixel modification amplitude.

Denote the original image as X , the watermark as W , the watermark embedding strength as α , the watermarked image as X_{Emb} , the noise pool as $N = N_1, N_2, \dots, N_n$, the convolution operator as K , the deconvolution operator as K^{-1} , the activation function as σ , and the number of convolution layers as l , then the encoder RRE process can be expressed as:

$$X_{Emb} = X + \alpha \sum_{i=1}^n \omega_i E_i(X, W) \quad (7)$$

its pseudocode can be represented by Algorithm 1. The specific steps are as follows:

- Step 1: Load the original image X and randomly generate a watermark sequence W .
- Step 2: Convert X to RGB channels and preprocess W into a tensor matrix format.
- Step 3: Input X and W into the encoder RRE:
- There are a total of n layers of noise simulation. In each layer, convolution operations are performed on both X and W , with a total of l convolution operations.
 - Record the residual map R_i output by the encoder at each noise simulation layer.
 - Superimpose the residual map R_i onto X to obtain the watermarked image, and apply noise attacks to this watermarked image.
 - Decode the watermarked image after noise attacks, and extract the watermark W_{temp} under each noise simulation layer.
 - Compute the robustness weight ω_i under each noise simulation layer, which serves as the weight for the residual R_i .
- Step 4: Perform weighted fusion of the residual maps obtained from the n layers of noise simulation, and superimpose the result onto X to obtain the final watermarked image X_{Emb} .

To ensure that the encoder E_i effectively couples watermark features with image features, we concatenate the watermark along the third-dimensional channel of the feature map at each convolutional layer. We adopt a "Concatenation" approach to splice the watermark onto the channels along the third dimension, and then feed the result into the next convolutional layer for feature extraction. Similarly, after each subsequent convolutional layer, we continue to use "Concatenation" to splice the watermark onto the channels along the third dimension, and this process is repeated until the final output layer. This process is represented by a mapping function:

$$f_E^l : \mathbb{R}^{m_l \times n_l \times c_l} \rightarrow \mathbb{R}^{m_l \times n_l \times c_l + c} \quad (8)$$

where $m_l \times n_l \times c_l$ is the shape of l -level feature map, c is the number of channels of the watermark tensor. Since the shape of the feature map varies across layers in encoder E_i , the decoder uses a corresponding deconvolution structure to locate and extract the watermark features. Additionally, the convolution kernel size in each layer of the decoder is symmetrically aligned with that of the encoder.

Algorithm 1. Encoder RRE.

Input: Original image X , watermark W
Output: Watermarked image X_{Emb}

- 1 initialization: $x_0^{(0)} \leftarrow X$
- 2 $W_{temp} \leftarrow 0$, $R \leftarrow \vec{0}$, $\omega \leftarrow \vec{0}$
- 3 **for** $i=1:n$ **do**
- 4 $x_0^{(i)} = X$
- 5 **for** $j=1:l$ **do**
- 6 $x_j^{(i)} \leftarrow \sigma(K \circ x_{j-1}^{(i)} \odot W)$
- 7 **end**
- 8 $R \leftarrow x_j^{(i)}$
- 9 $x_0^{(i)} \leftarrow N_i(R_i + X)$
- 10 **for** $j=1:l$ **do**
- 11 $x_j^{(i)} \leftarrow \sigma(K^{-1} \circ x_{j-1}^{(i)})$
- 12 $W_{temp} \leftarrow x_j^{(i)}(:, :, -c :)$
- 13 $x_j^{(i)} \leftarrow x_j^{(i)}(:, :, -c :)$
- 14 **end**
- 15 $\omega \leftarrow Acc(W_{temp}, W)$
- 16 **end**
- 17 $X_{Emb} \leftarrow X + \alpha \sum_{i=1}^n \omega_i R_i$

C. Decoder

The main function of decoder is watermark extraction. In fact, in the decoder, different outputs can be designed according to the needs of downstream tasks, such as image recovery task. In order to extract the watermark, the design of the decoder has a certain relationship with the encoder. In other words, the decoder needs to know where the watermark is embedded, that is, the feature space, so that it can quickly locate the watermark feature space for precise extraction. According to the design of the encoder, we implement a deconvolution structure in the decoder, which decodes the watermarked image X_{Emb} , reconstructs the encoder's feature approximations and decouples the watermark features. Since each convolution layer of encoder is associated with watermark embedding, the features obtained by deconvolution of each layer can be decoupled from the watermark features:

$$f_D^l : \mathbb{R}^{m_l \times n_l \times c_l + c} \rightarrow \mathbb{R}^{m_l \times n_l \times c} \quad (9)$$

We weighted the decoupled watermark features and input them into a fixed-length one-dimensional vector with the sigmoid activation function to yield the extracted watermark W_{Ext} :

$$W_{Ext} = Sigmoid(Dense(Flatten(W_{temp}))) \quad (10)$$

where the Flatten layer is the flattening operation, which is used to convert the tensor into a one-dimensional vector. The Dense layer is a linear layer that can map one-dimensional vectors to a fixed length. W_{temp} is the extracted watermark tensor. The process of the decoder can be represented by Algorithm 2. The specific steps are as follows:

- Step 1: Load the watermarked image X_{Emb} .
- Step 2: Convert X_{Emb} to RGB channels.
- Step 3: Input X_{Emb} into the decoder D :

Algorithm 2. Decoder D .

Input: Watermarked image X_{Emb}
Output: Watermark W_{Ext}

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1 initialization:  $x_0^{(0)} \leftarrow X_{Emb}$ 
2  $W_{temp} \leftarrow 0$ 
3 for  $j=1:l$  do
4    $x_j^{(i)} \leftarrow \sigma(K^{-1} \circ x_{j-1}^{(i)})$ 
5    $W_{temp} \leftarrow x_j^{(i)}(:, :, -c:)$ 
6    $x_j^{(i)} \leftarrow x_j^{(i)}(:, :, -c)$ 
7 end
8  $W_{Ext} \leftarrow \text{sigmoid}(\text{Dense}(\text{Flatten}(W_{temp})))$ 

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- a) Perform deconvolution operations on X_{Emb} , with a total of l operations.
- bt) Extract watermark features from the feature maps at each convolution step.

Step 4: Fuse all the extracted watermark features.

Step 5: Apply the Sigmoid activation function and generate the extracted watermark W_{Ext} of length L .

D. Loss Function

In order to take into account the visual quality and robustness of the watermarked image, the loss function consists of two parts, namely, image loss and watermark cross-entropy loss. For the encoder RRE, the optimization is mainly carried out by the loss of $\mathcal{L}_1$, $\mathcal{L}_2$ and $\mathcal{L}_3$. Where $\mathcal{L}_1$ is the mean square error (MSE) [48], [49] loss of watermarked image X_{Emb} and original image X , which can be expressed as:

$$\begin{aligned}\mathcal{L}_1 &= \frac{1}{m \times n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} (X_{Emb}(i, j) - X(i, j))^2 \\ &= \frac{1}{m \times n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} R^2\end{aligned}\quad (11)$$

$\mathcal{L}_2$ is the imperceptible loss of watermarked image, which is mainly optimized by PSNR [50]:

$$\mathcal{L}_2 = \frac{1}{PSNR}\quad (12)$$

$\mathcal{L}_3$ is composed of the watermark cross-entropy loss of n noise types, which can be expressed as:

$$\begin{aligned}\mathcal{L}_3 &= \sum_{i=1}^n \mathcal{L}_{N_i} \\ &= \frac{1}{L} \sum_{i=1}^n \sum_{j=0}^{L-1} -w_{i,j} \log(w'_{i,j}) - (1 - w_{i,j}) \log(1 - w'_{i,j})\end{aligned}\quad (13)$$

where L is the length of the watermark W , and $w'_{i,j}$ represents the watermark bit extracted at the j -th position under the i -th type noise N_i , whose value is 0 or 1. $\mathcal{L}_{N_i}$ serves as the weight ω_i of each robust residual R_i , which is dynamically optimized with the number of iterations.

The decoder is optimized by loss of $\mathcal{L}_4$. The $\mathcal{L}_4$ is the global watermark cross-entropy loss, which measures the similarity between the decoder's extracted watermark W_{Ext} and the original watermark W :

$$\mathcal{L}_4 = \frac{1}{L} \sum_{j=0}^{L-1} -w_j \log(w'_j) - (1 - w_j) \log(1 - w'_j)\quad (14)$$

where w'_j represents the watermark extracted by the decoder D at the j -th position, and its value is 0 or 1.

Therefore, the final loss function of the network is:

$$\mathcal{L}_{Total} = \lambda_1 \mathcal{L}_1 + \lambda_2 \mathcal{L}_2 + \lambda_3 \mathcal{L}_3 + \lambda_4 \mathcal{L}_4\quad (15)$$

where λ_1 , λ_2 , λ_3 and λ_4 are the corresponding loss weights, whose initial values are (1.5, 1.0, 1.0, 1.0).

IV. EXPERIMENTS

To validate the generation quality, robustness, and generalizability, we conducted extensive experiments across multiple datasets under diverse noise conditions. Additionally, we performed ablation studies to assess framework components and highlighted its advantages through comparative analyses.

A. Experimental Setup

1) *Dataset*: The datasets include BOSSBase [51], Mirflickr [52] and COCO [53] datasets. Among them, For the training set, we randomly selected 2,000 color images from the Mirflickr database. The test set, consists of 100 randomly selected images from each of the BOSSBase (grayscale images), Mirflickr (color images), and COCO (color images) databases. Furthermore, since the input image size required by the proposed method is fixed at 400×400 , the resolution of all images is scaled to 400×400 before experimentation.

2) *Evaluation Metric*: The evaluation metrics focus primarily on visual quality and robustness. To assess visual quality, we use two metrics: average peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) [50]. Higher values for these metrics indicate better visual quality of the watermarked images. For robustness, we measure the watermark extraction accuracy under various types of attacks using average bit accuracy (ACC). Additionally, tests are conducted on multiple test sets to evaluate generalization performance.

3) *Implementation Detail*: The experiments are conducted on a Win10 operating system with an Intel(R) Xeon(R) Gold 6161 CPU @ 2.20 GHz and 128 GB of RAM. The code is implemented in Python 3.6. For training hyperparameter settings, the optimizer used is Adam with a default learning rate (lr) of $1e-4$, a batch size of 1, and 140,000 training steps. The convolution kernel size is set to 3×3 , the stride to 1, and the watermark length to 100. Among them, the watermark information corresponding to each image in both the training set and the test set is randomly generated by a random function. The randomly generated watermark is a bit sequence of length 100, which is fed into the encoder together with the image as the original input data.

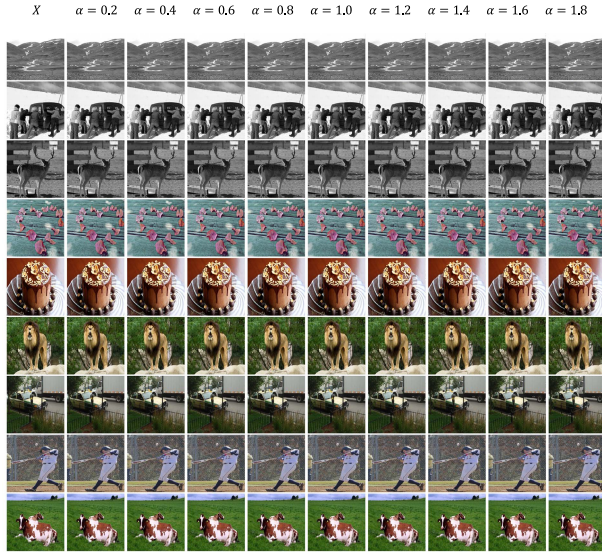


Fig. 6. Partial watermarked image X_{Emb} generated with different embedding strengths. The first three rows of images are from the BOSSBase database, the middle three rows are from the Mirflickr database, and the last three rows are from the COCO database.

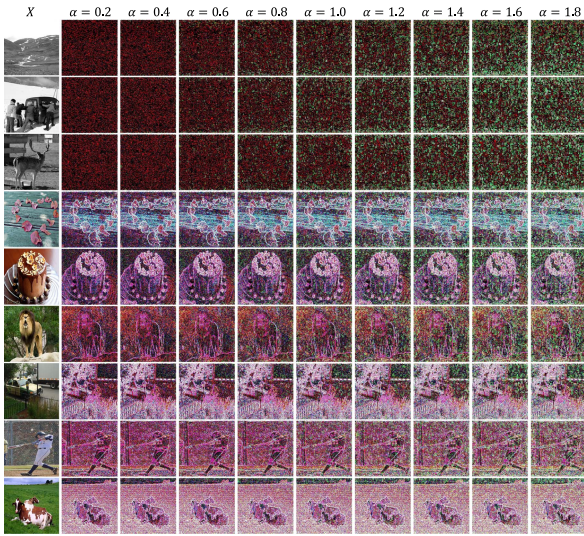


Fig. 7. Visualization of partial residual image $\|R\|$, where $\|R\| = \|X_{Emb} - X\|$. Since the pixel value r_{ij} ($0 \leq i \leq m, 0 \leq j \leq n$) of residual $\|R\|$ is relatively small, in order to be able to visualize, we need to binarize the pixel r_{ij} , that is, when the pixel $r_{ij} > 1, r_{ij} = 255$, otherwise $r_{ij} = 0$. Therefore, for the residual graph, the white region represents the strongly embedded watermark region, and the black region represents the weakly embedded watermark region.

B. Visual Quality

During network training, the embedding strength is set to $\alpha = 1.0$. The parameter α controls the balance between visual quality and robustness and can be adjusted based on application requirements. Fig. 6 displays sample watermarked images produced with different embedding strengths, while Fig. 7 shows the corresponding residual images for the watermarked image X_{Emb} . In the residual images, white areas indicate regions with a strong watermark embedding, whereas black areas represent weakly embedded regions. Furthermore, the proposed method

demonstrates consistently high PSNR and SSIM performance across the three test datasets, as shown in Table I. Notably, PSNR and SSIM values remain stable on cross-dataset evaluations with BOSSBase and COCO. When the embedding strength $\alpha = 1.0$, PSNR exceeds 40.0 dB, and SSIM exceeds 0.992, confirming the method's high-quality generation and generalization.

C. Robustness Under Visible Noise Attacks

While lower embedding strength can improve visual quality, it may weaken the watermark's resilience to noise attacks. Ideally, the watermarking method should maintain robustness even at lower embedding strengths. Therefore, in this section, we mainly test the robustness under different embedding strengths. For robustness, the noises tested include JPEG compression, Crop, Cropout, Dropout, Gaussian noise, Gaussian filter, Identity and Scaling, and their parameters are shown in Table II. To conserve space, only representative results are presented for embedding strengths of 0.2, 0.4, 0.6, 0.8, 1.0, and 1.8. Additional results can be verified in a demo that we will open source soon.

As can be seen from Table III, In the case of lower embedding strength, the robustness will be affected to some extent. The robustness is greatly reduced on JPEG compression and Gaussian noise attacks, but still maintains a certain robustness on other noises. According to the proposed watermark embedding method $X_{Emb} = X + R$, we know that when the added noise has a large effect on the residual R , the ability of the residual R to represent 0 and 1 will be weakened. In addition, low embedding strength will increase the risk of modification of residual R , which will further reduce its ability to represent 0 or 1. Therefore, in practical applications, if we can know the noise type and its approximate noise intensity, then we can train a robust watermarking network specifically for this scene.

In Table III, we actually list the attack results of two Gaussian noise parameters, where the variance of 0.02 is the maximum simulation variance set in the network training, and 0.1 is outside the range of the simulation parameter. However, from the results, the effect of robustness is very small. This phenomenon also shows that the residual R has good robustness, even for the attack parameters that are not simulated. Finally, we show some watermarked images after noise attacks, as illustrated in Fig. 8. These images simulate real-world distortions that may occur during image transmission, storage, or manipulation. Although the visual quality of the watermarked images is degraded by these attacks, they serve as the basis for evaluating the robustness of watermark extraction.

D. Robustness Under Invisible Noise Attacks

This section evaluates the robustness of the proposed method against invisible noise attacks, where such perturbations are not involved during training. In principle, the method is not expected to perform well under these unseen distortions. The considered attack types include perspective warping (PW), color transformation (CT), gradient-based attacks such as projected gradient descent (PGD) [54] and carlini & wagner (CW) [55], as well as median blur (MB). PW simulates lens distortion or physical deformation by generating a perspective transformation

TABLE I
THE PSNR AND SSIM OF THREE TEST SETS AT DIFFERENT EMBEDDING STRENGTHS

α		0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8
BOSSBase	PSNR $\uparrow$	50.3979	47.6872	45.0947	43.0337	41.2928	39.8175	38.5758	37.5542	36.6406
	SSIM $\uparrow$	0.9980	0.9972	0.9959	0.9942	0.9922	0.9900	0.9876	0.9850	0.9822
Mirflickr	PSNR $\uparrow$	50.4159	47.6620	45.0900	42.9281	41.2138	39.7190	38.4988	37.4530	36.5254
	SSIM $\uparrow$	0.9977	0.9970	0.9957	0.9941	0.9921	0.9900	0.9874	0.9849	0.9822
COCO	PSNR $\uparrow$	50.1685	47.2623	44.6639	42.4873	40.6964	39.2425	38.0277	36.9384	36.0074
	SSIM $\uparrow$	0.9980	0.9972	0.9960	0.9945	0.9926	0.9906	0.9884	0.9859	0.9833

TABLE II
ATTACK TYPES AND PARAMETER DESCRIPTION

Attack types	Parameter description
Identity	Without noise attack.
Gaussian noise	The parameter standard deviation σ ranges from $[0, 0.1]$.
Gaussian filtering	The filter kernel k takes the values of 3×3 , 5×5 and 7×7 .
JPEG	The JPEG compression factor QF ranges from $[50, 100]$.
Crop	The Crop factor range is $[0, 0.03]$.
Scaling	The scaling factor ranges from $[0, 0.5]$.
Dropout	The probability p ranges from $[0, 1]$, which represents that some pixels of watermarked image are randomly selected by probability p to be retained, and the rest are filled by original image.
Cropout	The probability p ranges from $[0, 1]$, which represents that a random square pixels of watermarked image are randomly selected by probability p to be retained, and the rest are filled by original image.



Fig. 8. Watermarked images and their corresponding ACC under various attacks.

matrix through random displacement of the image's four corner points, with the distortion level controlled by a parameter—the larger the value, the stronger the warping. CT introduces random perturbations in the HSV color space, including hue, saturation, and value components, where the strength of perturbation is similarly governed by a parameter indicating the severity of the color shift. For PGD, the parameters include the maximum perturbation magnitude ϵ , step size α , and the number of iterations n , where ϵ is set to 0.01, 0.05, and 0.1, α is 20% of the ϵ , and n

is fixed at 10. The CW attack uses confidence, learning rate, and number of iterations n as parameters, with the confidence and learning rate fixed at $1e-4$ and 0.01 respectively, and n set to 50, 100, and 200. For MB, the main parameter is the filter kernel size k , which takes values of 3, 5, and 7.

Table V presents the robustness of the proposed method under invisible noise attacks. The results show that such unseen perturbations significantly impact the method's performance. For PW attack, the method retains a certain degree of robustness only at lower attack intensities. In contrast, for CT attack, the method maintains high robustness across all tested levels. For PGD and CW attacks, the method also demonstrates strong resilience. In the case of MB, the method performs well at low filter strengths, but its robustness degrades as the filter size increases. These findings suggest that the proposed method possesses inherent robustness against invisible attacks. We attribute this to the network's ability to learn strongly robust residuals. Such residuals are known to be linearly additive across multiple types of noise and exhibit redundancy in real-world scenarios—meaning that the robust residual for one type of noise may partially or entirely include the residuals of other noise types. Based on this observation, we argue that the strongly robust residuals learned by our method implicitly capture components corresponding to unseen attack types, thereby enabling generalization to invisible distortions. To further strengthen robustness in practical applications, we recommend incorporating a range of potential noise perturbations into the encoder during training, thereby guiding the network to learn even more generalized robust residuals and minimizing vulnerability to unforeseen adversarial conditions. Finally, we show some watermarked images after invisible noise attacks, as illustrated in Figs. 10 and 11.

E. Ablation Experiment

1) *Noise Layer*: According to the proposed watermarking method, the residual R generated by the encoder RRE has very strong robustness, so there is no need to add noise layer for image distortion simulation. In other words, the proposed method no longer requires the noise layer N in the E-N-D structure, but is directly connected to the decoder D . In order to ablate the necessity of noise layer N , we tested the performance of the E-N-D structure, that is, after the encoder RRE outputs the robust residual R , it is added to the original image X to produce the watermarked image X_{Emb} , and then connected to the noise layer N .

Fig. 9(a) shows the Epoch-Loss diagram of the network trained with noise layer N . We can find that the optimization

TABLE III
THE ROBUSTNESS OF DIFFERENT EMBEDDING STRENGTHS

Dataset	α	Identity	JPEG (QF=50)	Crop ($p=0.03$)	Cropout ($p=0.9$)	Dropout ($p=0.9$)	Gaussian noise ($\sigma=0.02$)	Gaussian noise ($\sigma=0.1$)	Gaussian filter ($k=7$)	Scaling ($p=0.5$)
BOSSBase	0.2	96.16%	72.63%	92.79%	95.88%	95.96%	90.96%	69.43%	95.91%	89.60%
	0.4	99.69%	92.25%	93.16%	99.77%	99.66%	98.78%	81.47%	99.75%	96.27%
	0.6	100.0%	98.54%	95.31%	100.0%	99.99%	99.87%	90.28%	100.0%	99.16%
	0.8	100.0%	99.81%	97.85%	100.0%	100.0%	99.99%	95.91%	100.0%	99.90%
	1.0	100.0%	99.90%	98.90%	100.0%	100.0%	100.0%	98.09%	100.0%	99.97%
	1.8	100.0%	99.99%	99.47%	100.0%	100.0%	100.0%	99.77%	99.99%	100.0%
Mirflickr	0.2	95.45%	71.44%	89.12%	95.44%	95.44%	90.68%	66.61%	95.54%	88.78%
	0.4	99.29%	89.98%	93.40%	99.35%	99.37%	98.62%	79.49%	99.16%	95.81%
	0.6	99.90%	97.49%	96.73%	99.94%	99.92%	99.88%	87.91%	99.98%	98.79%
	0.8	100.0%	99.45%	97.85%	100.0%	99.99%	99.98%	93.21%	100.0%	99.77%
	1.0	100.0%	99.89%	98.40%	100.0%	100.0%	100.0%	95.93%	100.0%	99.89%
	1.8	100.0%	99.99%	99.36%	100.0%	100.0%	100.0%	99.09%	100.0%	99.97%
COCO	0.2	93.16%	71.68%	88.05%	92.98%	93.03%	88.57%	68.68%	93.31%	86.27%
	0.4	99.08%	90.10%	92.25%	99.16%	99.16%	98.32%	81.89%	99.07%	94.54%
	0.6	99.98%	97.70%	96.64%	99.98%	99.96%	99.88%	90.85%	99.90%	98.17%
	0.8	100.0%	99.64%	96.50%	100.0%	99.99%	99.98%	95.65%	100.0%	99.58%
	1.0	100.0%	99.97%	97.63%	100.0%	100.0%	99.99%	97.59%	100.0%	99.83%
	1.8	100.0%	99.96%	99.00%	100.0%	100.0%	100.0%	99.56%	100.0%	100.0%

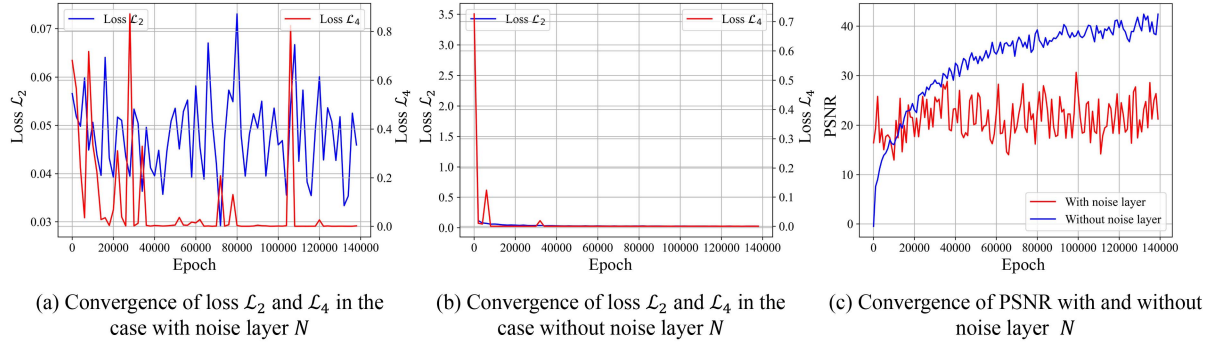


Fig. 9. The ablation of noise layer N in E-N-D structure. (a) is the Epoch-Loss diagram of network training with noise layer N . (b) is the Epoch-Loss diagram of network training without noise layer N . (c) is the convergence diagram of PSNR during network training in both cases.

TABLE IV
THE PERFORMANCE OF THE NETWORK IN THREE TEST SETS WITH(W) AND WITHOUT(WO) NOISE LAYER N

Dataset	PSNR(w/wo) $\uparrow$	SSIM(w/wo) $\uparrow$	Identity(w/wo) $\uparrow$
BOSSBase	19.7027/ 41.2928	0.3215/ 0.9922	100.0%/100.0%
Mirflickr	19.1826/ 41.2138	0.3542/ 0.9921	100.0%/100.0%
COCO	19.2459/ 40.6964	0.3614/ 0.9926	100.0%/100.0%

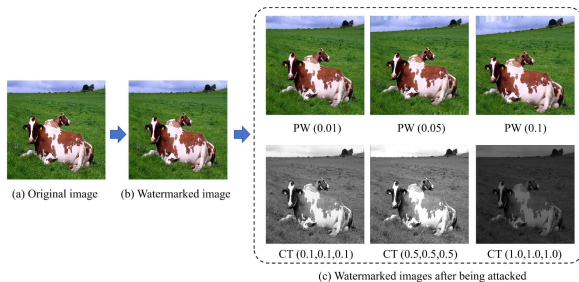


Fig. 10. Watermarked images attacked by invisible noises PW and CT.

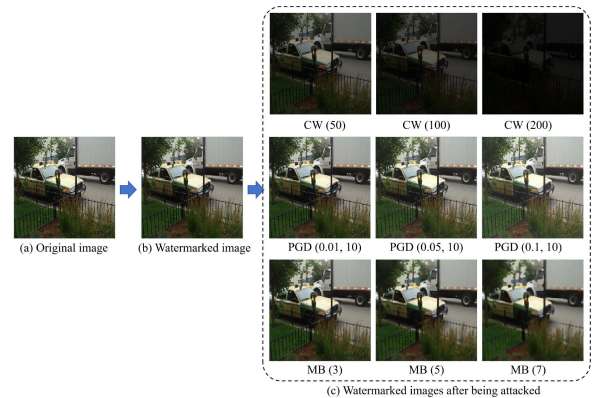


Fig. 11. Watermarked images attacked by invisible noises CW, PGD and MB.

of the network is biased towards robustness (corresponding loss $\mathcal{L}_4$) and ignores the visual quality (corresponding loss $\mathcal{L}_2$). In fact, according to the proposed method, we know that the residual R generated by the encoder RRE is robust. If a noise layer

TABLE V
THE ROBUSTNESS OF THE THREE DATASETS AGAINST INVISIBLE NOISE
ATTACKS (EMBEDDING STRENGTH IS 1.0)

Attack Types	Attack parameters	BOSSBase	Mirflickr	COCO
PW	0.01	62.76%	59.53%	64.62%
	0.05	51.04%	51.27%	52.43%
	0.1	50.68%	50.71%	50.70%
CT	(0.1,0.0,0.0)	99.65%	99.66%	99.63%
	(0.5,0.0,0.0)	99.66%	99.51%	99.64%
	(1.0,0.0,0.0)	99.54%	99.63%	99.55%
	(0.0,0.1,0.0)	99.62%	99.59%	99.54%
	(0.0,0.5,0.0)	99.60%	99.66%	99.63%
	(0.0,1.0,0.0)	99.58%	99.53%	99.66%
	(0.0,0.0,0.1)	99.57%	99.57%	99.58%
	(0.0,0.0,0.5)	99.27%	98.54%	98.87%
	(0.0,0.0,1.0)	99.16%	97.43%	97.06%
	(0.1,0.1,0.1)	99.67%	99.53%	99.64%
	(0.5,0.5,0.5)	99.58%	99.09%	99.21%
	(1.0,1.0,1.0)	98.52%	98.62%	97.56%
CW	50	99.91%	99.64%	99.86%
	100	87.15%	86.47%	95.44%
	200	34.32%	21.33%	24.17%
PGD	0.01	100.0%	100.0%	100.0%
	0.05	100.0%	100.0%	100.0%
	0.1	100.0%	100.0%	100.0%
MB	3	100.0%	100.0%	100.0%
	5	93.47%	92.27%	92.00%
	7	69.36%	63.40%	64.25%

D is added, the value of the robust residual R will be optimized again. Since the noise layer is mainly optimized for robustness, it will further strengthen the robustness of the residual R , resulting in a decrease in visual quality. In other words, the robust residual R is modified again, disrupting the balance between robustness and visual quality. In contrast, without the presence of noise layer N , the network achieves stable convergence in optimizing visual quality and watermark cross-entropy, as illustrated in Fig. 9(b). Fig. 9(c) shows the optimization of PSNR during the corresponding training of two kinds of networks. It can be seen that without the noise layer N , the network can be well optimized in terms of visual quality and robustness. However, in the case of containing noise layer N , the visual quality process oscillation optimization cannot converge well. At the same time, we tested the performance of both networks on three test sets, as shown in Table IV. It can be found that the network can achieve good performance in terms of robustness, but in the case of containing noise layer, the optimization of visual quality will be affected, resulting in the failure of convergence. Therefore, after the encoder RRE generates the robust residual R , it is no longer necessary to add the noise layer N to simulate the watermarked image distortion.

2) *Residual Weight*: For robustness, it is mainly reflected in the robust residual R generated by the new encoder RRE. However, a robust watermarking network with strong applicability should be able to resist all noise types. According to formula (3), the robustness corresponding to each noise has a corresponding robust weight, and this weight will directly affect the robustness of the residual R . According to normal thinking, if the watermark extraction accuracy corresponding to noise N_i is higher, or the

TABLE VI
THE VISUAL QUALITY OF THE NETWORKS CORRESPONDING TO THE TWO
RESIDUAL WEIGHTS

Dataset	ω_i	PSNR $\uparrow$	SSIM $\uparrow$	Identity $\uparrow$
BOSSBase	$\mathcal{L}_{N_i}$	41.2928	0.9922	100.0%
	$1 - \mathcal{L}_{N_i}$	28.9285	0.9305	99.92%
Mirflickr	$\mathcal{L}_{N_i}$	41.2138	0.9921	100.0%
	$1 - \mathcal{L}_{N_i}$	28.9489	0.9332	99.90%
COCO	$\mathcal{L}_{N_i}$	40.6964	0.9926	100.0%
	$1 - \mathcal{L}_{N_i}$	28.4150	0.9391	99.83%



Fig. 12. Visualization of watermarked image X_{Emb} and residual image R generated by different robust residual weights. Where $X_{Emb} \sim \mathcal{L}_{N_i}$ is the watermarked image generated by the network trained with $\mathcal{L}_{N_i}$ as robust residual weight, and $R \sim \mathcal{L}_{N_i}$ is its corresponding residual. $X_{Emb} \sim 1 - \mathcal{L}_{N_i}$ is the watermarked image generated by the network trained with $1 - \mathcal{L}_{N_i}$ as robust residual weight, and $R \sim 1 - \mathcal{L}_{N_i}$ is its corresponding residual.

cross-entropy $\mathcal{L}_{N_i}$ is lower, then its corresponding residual R_i should have higher robustness, so R_i should be assigned a higher weight ω_i . However, in the experiment, we found that for simple noise types, such as Gaussian noise, Identity or other attack types, the network is easy to learn and the accuracy of watermark extraction is high, but their corresponding residual is not very robust. In other words, the value of the residual is small, and the information content is insufficient. Therefore, according to the offline experiments, we come to the conclusion that the robust residual R_i is given a higher weight ω_i for the complex noise type N_i . For the simple noise type N_i , the robust residual R_i is given a lower weight ω_i . Here, we use cross-entropy $\mathcal{L}_{N_i}$ to measure the complexity of the noise, and directly as the weight ω_i . Then, when the cross-entropy $\mathcal{L}_{N_i}$ is small, the watermark accuracy is high, otherwise the opposite.

Table VI shows the experimental results using $\mathcal{L}_{N_i}$ and $1 - \mathcal{L}_{N_i}$ as robust residual weights. From the experimental results, it is true that for complex noise types, it should be given a higher robust weight. In fact, it is not difficult to find that for complex noise type N_i , the generated residual R_i often has a larger value, that is, it contains more information, and may contain the corresponding residual information of simple noise type. Therefore, the residuals caused by noise are not actually independent, they have a certain redundancy, but this does not affect the residuals R , but only affects the weight of the noise type.

At the same time, it can be seen from Fig. 12 that the visual effects of watermarked images and residual images generated

TABLE VII
THE INFLUENCE OF LOSS WEIGHTS λ_1 , λ_2 , λ_3 AND λ_4 ON THE PERFORMANCE OF THE ALGORITHM (EMBEDDING STRENGTH IS 1.0)

$(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$	PSNR			SSIM			JPEG (QF=50)		
	BOSSBase	Mirflickr	COCO	BOSSBase	Mirflickr	COCO	BOSSBase	Mirflickr	COCO
(1.0, 1.0, 1.0, 1.0)	38.8743	38.7805	38.3946	0.9888	0.9876	0.9893	99.89%	99.76%	99.85%
(1.5, 1.0, 1.0, 1.0)	41.2928	41.2138	40.6964	0.9922	0.9921	0.9926	99.90%	99.89%	99.97%
(1.0, 1.5, 1.0, 1.0)	42.0568	42.2799	41.7909	0.9932	0.9928	0.9934	99.77%	99.15%	99.62%
(1.0, 1.0, 1.5, 1.0)	38.9423	38.7128	38.5237	0.9873	0.9857	0.9876	99.96%	99.92%	99.94%
(1.0, 1.0, 1.0, 1.5)	37.2952	37.0723	36.7734	0.9869	0.9864	0.9876	99.96%	99.95%	99.98%
(1.5, 1.5, 1.0, 1.0)	40.3693	40.3954	39.8298	0.9928	0.9925	0.9931	99.96%	99.90%	99.91%
(1.0, 1.0, 1.5, 1.5)	35.4255	35.5341	35.0000	0.9853	0.9856	0.9865	100.0%	100.0%	100.0%

by two networks trained with different weights. Among them, the white region in the residual graph represents the strongly embedded watermark region, and the black region represents the weakly embedded region. For the watermarked image with weight $1 - \mathcal{L}_{N_i}$, although the difference between the watermarked image and the original image cannot be seen by the naked eye, we can see some modification traces after enlarging the watermarked image. However, for the watermarked image with weight $\mathcal{L}_{N_i}$, no obvious modification traces can be seen even if the watermarked image is enlarged.

3) *Loss Weight*: To analyze the effect of each loss weight λ_i ($i=1,2,3,4$) in the total loss function $\mathcal{L}_{Total}$, we conduct extensive experiments by adjusting λ_1 , λ_2 , λ_3 and λ_4 . We note that λ_1 and λ_2 mainly affect the visual quality loss of the watermarked image, while λ_3 and λ_4 relate to the distortion and preservation of the embedded watermark information. Therefore, we test both average-weighted and non-average-weighted settings across 7 groups of experiments. The results, as shown in Table VII, demonstrate that the parameters λ_i ($i=1,2,3,4$) significantly influence both imperceptibility and robustness. When λ_1 and λ_2 increase, the network focuses more on optimizing visual quality, producing watermarked images with higher fidelity (as indicated by improvements in PSNR and SSIM). However, this comes at the cost of reduced robustness. Conversely, when λ_3 and λ_4 increase, the network places more emphasis on preserving the watermark, which enhances robustness considerably, but may degrade the visual quality of the generated watermarked images. In summary, we find that the selection of parameters λ_i ($i=1,2,3,4$) directly affects the trade-off between visual imperceptibility and robustness. By properly adjusting λ_1 and λ_2 , we can achieve a better balance between these two critical aspects.

F. Comparative Experiment

From the previous experiments, we established that an effective watermarking method should demonstrate high generation quality, robustness, and generalization. If the generalization is good, then the comparison experiment should be independent of the test set. However, because the input image size and embedded watermark length of each method are different, it is difficult to achieve the fairness of the comparison to some extent. Nonetheless, we try to keep the comparison methods under the same test set for testing. In this section, we compare our method with the HiDDeN [28], StegaStamp [29], MBRS [30],

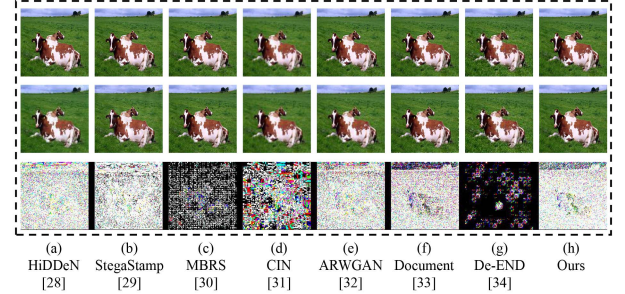


Fig. 13. Visualization of watermarked image generated by different watermarking methods. Among them, the first row is the input image X , the second row is the watermarked image X_{Emb} , and the third row is the corresponding residual graph R .

CIN [31], ARWGAN [32], Document [33], De-END [34] and ST-DCN [35], all of which are based on the E-N-D structure. We used their open source pre-trained model for testing, and for methods without open source pre-trained models, we retrained the network for replication. Additionally, we will also directly use the experimental data in the original paper (there may be different attack parameters tested) for presentation, and these data are marked with asterisks.

1) *Visual Quality Analysis*: Due to differences in input image sizes and watermark lengths across watermarking methods, the resolutions of generated watermarked images vary, impacting the fairness of visual quality and robustness comparisons. In terms of visual quality, De-END achieves high PSNR since it is trained exclusively with the Identity transformation. Our method achieves a PSNR exceeding 40.0 dB, ranking just below MBRS and CIN. And for SSIM, our method outperforms the other approaches overall. Although our method yields a lower PSNR compared to MBRS and CIN, it achieves higher SSIM scores. More importantly, our method outperforms both MBRS and CIN in terms of robustness across various noise attacks, including JPEG compression, Cropout, Dropout, Gaussian noise, and Scaling attack (See Table VIII). Therefore, from a comprehensive perspective, our approach demonstrates a clear advantage over the compared methods. An example of watermarked images generated by each method is shown in Fig. 13.

To examine how resolution affects the PSNR and SSIM indicators, we conduct a simple experiment, that is, after scaling the original image and the generated watermarked image, calculate their PSNR and SSIM again. From Table IX, results show that lower resolutions yield higher PSNR and SSIM values. In fact,

TABLE VIII
THE ROBUSTNESS OF DIFFERENT WATERMARKING METHODS

Dataset	Method	$m \times n$	L	PSNR↑	SSIM↑	Identity	JPEG (QF=50)	Crop ($p=0.03$)	Cropout ($p=0.9$)	Dropout ($p=0.9$)	Gaussian noise ($\sigma=0.02$)	Gaussian noise ($\sigma=0.1$)	Gaussian filter ($k=7$)	Scaling ($p=0.5$)
BOSSBase	HiDDeN [28]	128×128	30	35.9400	0.9704	97.00%	87.27%	75.70%	92.77%	98.47%	98.76%	91.47%	88.77%	77.53%
	StegaStamp [29]	400×400	100	30.0683	0.9100	99.99%	99.97%	99.75%	99.60%	99.93%	99.92%	98.28%	99.97%	99.91%
	MBRS [30]	256×256	256	42.9745	0.9882	100.0%	99.87%	49.24%	94.42%	100.0%	78.99%	60.86%	99.30%	50.18%
	CIN [31]	128×128	30	43.2185	0.9837	100.0%	99.60%	100.0%	100.0%	96.40%	100.0%	100.0%	100.0%	99.30%
	ARWGAN [32]	128×128	30	31.7361	0.9590	95.87%	85.93%	97.20%	99.17%	98.93%	93.09%	78.23%	99.30%	97.70%
	Document [33]	400×400	100	32.2991	0.9929	99.96%	99.98%	53.19%	99.97%	99.99%	99.95%	99.45%	99.89%	99.96%
	De-END [34]	128×128	64	49.1264	0.9805	100.0%	65.18%	—	98.32%	100.0%	98.37%	86.56%	100.0%	—
	ST-DCN* [35]	224×224	196	35.9730	0.9920	100.0%	99.93%(40)	96.75%(0.2)	96.87%(0.5)	95.44%(0.7)	98.09%(0.04)	—	100.0%	99.38%(0.4)
	Ours	400×400	100	41.2928	0.9922	100.0%	99.90%	98.90%	100.0%	100.0%	100.0%	98.09%	100.0%	99.97%
Mirflickr	HiDDeN [28]	128×128	30	32.5588	0.9373	93.75%	91.83%	75.37%	91.53%	97.50%	97.76%	91.60%	88.03%	78.23%
	StegaStamp [29]	400×400	100	29.3017	0.8903	99.93%	99.89%	99.61%	99.32%	99.85%	99.84%	97.95%	99.92%	99.84%
	MBRS [30]	256×256	256	41.9893	0.9877	100.0%	99.36%	49.78%	94.22%	100.0%	81.12%	64.14%	55.47%	49.93%
	CIN [31]	128×128	30	41.4366	0.9795	100.0%	99.67%	99.93%	100.0%	98.30%	100.0%	100.0%	100.0%	98.93%
	ARWGAN [32]	128×128	30	22.6466	0.8773	99.16%	90.27%	95.80%	98.87%	99.23%	94.50%	86.23%	99.13%	95.83%
	Document [33]	400×400	100	32.1330	0.9913	99.13%	99.16%	52.85%	99.22%	99.21%	98.83%	97.67%	95.45%	98.53%
	De-END [34]	128×128	64	48.7496	0.9510	98.48%	70.41%	—	97.56%	100.0%	91.34%	81.02%	99.90%	—
	ST-DCN* [35]	224×224	196	35.9730	0.9920	100.0%	99.93% (40)	96.75%(0.2)	96.87%(0.5)	95.44%(0.7)	98.09%(0.04)	—	100.0%	99.38%(0.4)
	Ours	400×400	100	41.2138	0.9921	100.0%	99.89%	98.40%	100.0%	100.0%	100.0%	95.93%	100.0%	99.89%
COCO	HiDDeN [28]	128×128	30	32.8929	0.9472	98.33%	92.73%	74.22%	92.23%	98.23%	97.93%	93.87%	89.93%	79.13%
	StegaStamp [29]	400×400	100	30.0683	0.9100	99.88%	99.82%	99.62%	99.18%	99.79%	99.88%	98.19%	99.88%	99.90%
	MBRS [30]	256×256	256	42.0076	0.9879	100.0%	99.62%	49.61%	93.96%	100.0%	79.57%	61.32%	100.0%	50.36%
	CIN [31]	128×128	30	41.9093	0.9827	100.0%	99.77%	99.97%	100.0%	97.40%	100.0%	100.0%	100.0%	99.27%
	ARWGAN [32]	128×128	30	31.7361	0.9590	99.20%	94.86%	96.23%	99.23%	99.06%	96.13%	87.50%	99.20%	96.30%
	Document [33]	400×400	100	32.2991	0.9929	99.96%	99.95%	53.19%	99.42%	99.51%	99.44%	98.35%	99.89%	99.42%
	De-END [34]	128×128	64	49.0125	0.9897	98.41%	70.36%	—	97.75%	100.0%	96.58%	87.98%	99.98%	—
	ST-DCN* [35]	224×224	196	35.9730	0.9920	100.0%	99.93%(40)	96.75%(0.2)	96.87%(0.5)	95.44%(0.7)	98.09%(0.04)	—	100.0%	99.38%(0.4)
	Ours	400×400	100	40.6964	0.9926	100.0%	99.97%	97.63%	100.0%	100.0%	100.0%	97.59%	100.0%	99.83%

TABLE IX
THE VISUAL QUALITY UNDER DIFFERENT RESOLUTIONS

Dataset		128×128	224×224	256×256	400×400
BOSSBase	PSNR↑	44.9698	43.3871	42.7859	41.2928
	SSIM↑	0.9965	0.9949	0.9943	0.9922
Mirflickr	PSNR↑	44.9535	43.2111	42.6146	41.2138
	SSIM↑	0.9966	0.9947	0.9942	0.9921
COCO	PSNR↑	44.5950	42.8211	42.1907	40.6964
	SSIM↑	0.9967	0.9950	0.9945	0.9926

according to the definition of the indicator, it is not difficult to find that in the case of unchanged embedded watermark capacity, when the resolution becomes smaller (that is, the m and n become smaller), PSNR and SSIM will be increased. Therefore, when resolution is consistent across methods, our proposed method achieves the highest watermarked image quality.

2) *Robustness Analysis*: In addition to the different input image sizes of each method, in fact, the length of each method embedded watermark is also different. Therefore, comparing the performance of each method is only a reference, and it is also necessary to combine various factors to comprehensively judge the advantages of the method. Like the framework proposed, its advantage is mainly reflected in the fact that it can directly generate robust residual R , without the need for noise layer N simulation. The residual R generated simultaneously is separable and has a linear combination of robust residuals of various noise types. As can be seen from Table VIII, when the embedding strength $\alpha = 1$, the robustness of our method achieves very good performance, and this robustness is achieved at a high generated visual quality. In general, robustness and visual quality are mutually restricted, when the visual quality of the watermarked image is poor, robustness is often very good. When the visual quality of watermarked images is high, the robustness is often poor. In the methods with image resolution of 400×400 and embedded watermark length $L = 100$, we can find that our method is better than other methods. Among all the watermarking methods, our method achieves the best

TABLE X
THE AVERAGE TEST TIME OF DIFFERENT WATERMARKING ALGORITHMS (SECONDS)

Methods	HiDDeN	StegaStamp	MBRS	CIN	ARWGAN	Document	De-END	Ours
BOSSBase	0.1611	0.2726	1.4261	1.2698	0.6057	0.2354	0.5007	0.5670
Mirflickr	0.1512	0.2766	1.3909	1.2652	0.5167	0.2403	0.5072	0.5565
COCO	0.1532	0.2745	1.4285	1.2521	0.5620	0.2595	0.5106	0.5601

performance in the robustness of some attack types. However, our method still has some shortcomings in some attack types, such as when the Gaussian noise variance is large, our method will have some disadvantages. So in the future, we plan to improve from adding noise, network design and other modules, such as adding more noise or increasing the attack strength, adding discriminators, attention mechanisms and so on.

3) *Time Complexity Analysis*: To analyze the time complexity of each watermarking algorithm, we evaluate the average testing time across three datasets. The average testing time refers to the total time required for embedding and extracting a watermark on a single image. For fairness, all evaluations are conducted under the Identity (no-attack) setting. Table X presents the average runtime of each algorithm on the three datasets. The results show that all watermarking algorithms exhibit very low embedding and extraction times. Among them, HiDDeN achieves the shortest runtime, while MBRS is the most time-consuming. Our method requires approximately 0.56 seconds on average for both embedding and extraction, which is faster than MBRS, CIN, and ARWGAN. Nonetheless, even without considering real-time constraints, all watermarking algorithms demonstrate excellent time efficiency, indicating strong practical value and applicability.

G. Security and Robustness Analysis

In the experiment, we first evaluate the imperceptibility of the watermarked images under different watermark embedding strengths. Next, with the embedding strength fixed at 1.0, we evaluate the robustness and security of the framework through experiments on visible and invisible noise attacks. Meanwhile,

we verify the rationality and feasibility of the framework via ablation studies. Finally, we further validate the effectiveness through comparative experiments. Experimental results demonstrate that even under invisible noise attacks, the method maintains high image quality and robustness, fully proving its security against noise attacks to a certain extent. Compared with existing schemes, the parallel approach adopted to optimize noise robustness not only improves the visual quality of the generated watermarked images but also significantly enhances resistance to various noise attacks. These results confirm that the proposed framework enhances robustness while preserving watermark imperceptibility, making it highly suitable for image security protection under signal processing and geometric attack environments, and at the same time, it provides new insights for the development of deep learning-based watermarking techniques.

V. CONCLUSION

This paper introduces a novel robust watermarking framework, LARW. Unlike the traditional E-N-D architecture, the LARW generates robust residuals directly through the redesigned encoder RRE, which are then added to the original image to generate the watermarked image. This approach eliminates the need to simulate image distortion, effectively removing the noise layer N in the E-N-D structure. The framework exhibits strong robustness, allowing for the integration of multiple noise types and the generation of robust, separable residuals. Experimental results validate the effectiveness of the proposed method, showing superior performance in both visual quality and resistance to common attack types. However, this paper still has limitations, such as the use of a fixed embedding strength and insufficient simulation of noise types. In future work, we plan to explore adaptive embedding strength mechanisms and evaluate the method on more diverse image domains and attack types. This adaptive mechanism could optimize the balance between imperceptibility, robustness, and capacity, making the watermarking process even more efficient and secure. These efforts can contribute significantly to the advancement of robust and trustworthy digital watermarking systems.

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